

C. One-time pad

Idea : use a key as long as the message

Example : Plaintext

JULIUS $\rightarrow$ KXNPVT

Plaintext (1 3 2 7 1 1) Ciphertext
key (random)

Unbreakable cryptosystem :

The plaintext is independent from the ciphertext, i.e.,
If you observe the ciphertext, then you cannot learn
any information about the plaintext.

Let's prove that the ciphertext and the plaintext are
independent.

Proof . To simplify the analysis (without loss of
generality) assume that the plaintext is a
sequence of bits and the key is a sequence
of bits

Plaintext: 0 1 1 0 1 Ciphertext: 1 1 1 1 0
Key : 1 0 0 1 1

modula 2 addition

	0	1
0	0	1
1	1	0

To further simplify, assume that the plaintext is a single bit

Plaintext: $M \in \{0, 1\}$ (one bit)

Key: $K \in \{0, 1\}$ (one random bit)

$$P[K=0] = P[K=1] = \frac{1}{2}$$

The key is independent from M

Encryption: $C = M + K \pmod{2}$ written as $C = M \oplus K$

Decryption: $M = C - K \pmod{2}$

$$\begin{aligned} \text{can also be written as } & C \oplus K \\ &= M \oplus \underbrace{K \oplus K}_{=0} \\ &= M \oplus 0 \\ &= M \end{aligned}$$

We want to show that M and C are independent

Review: let P_{MC} be the joint probability distribution of M and C .

$$\begin{array}{ll} P_{MC}(0,0) & P_{MC}(1,0) \\ P_{MC}(0,1) & P_{MC}(1,1) \end{array}$$

Independence of M and C means that
we have $P_{MC} = P_M \times P_C$, i.e.,

$$\left\{ \begin{array}{l} P_{MC}(0,0) = P_M(0) \times P_C(0) \\ P_{MC}(0,1) = P_M(0) \times P_C(1) \\ P_{MC}(1,0) = P_M(1) \times P_C(0) \\ P_{MC}(1,1) = P_M(1) \times P_C(1) \end{array} \right.$$

$$\begin{aligned} (1) P_{MC}(0,0) &= \underbrace{P_{C|M}(0|0)} P_M(0) \\ &= IP[C=0|M=0] \\ &= IP[M \oplus K=0|M=0] \\ &= IP[K=0|M=0] \\ &= IP[K=0] \quad (\text{because } M \text{ and } K \text{ are independent}) \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} (2) P_C(0) &= IP[C=0] = IP[M \oplus K=0] = IP[M \oplus M \oplus K=0 \oplus M] \\ &= IP[K=M] = \frac{1}{2} \end{aligned}$$

From (1) and (2), we have shown $P_{MC}(0,0) = P_C(0) \times P_M(0)$

We also have

$$\begin{aligned} P_{MC}(1,0) &= \underbrace{P_{C|M}(0|1)} P_M(1) \\ &= P[M \oplus K = 0 \mid M=1] \\ &= P[1 \oplus K = 0 \mid M=1] \\ &= P[K=1 \mid M=1] \\ &= P[K=1] \quad (\text{because } M \text{ and } K \\ &\quad \text{are independent}) \\ &= \frac{1}{2} \\ &= P_C(0) \end{aligned}$$

We also have

$$\begin{aligned} P_{MC}(1,1) &= \underbrace{P_{C|M}(1|1)} P_M(1) \\ &= P[M \oplus K = 1 \mid M=1] \\ &= P[1 \oplus K = 1 \mid M=1] \\ &= P[K=0 \mid M=1] \\ &= P[K=0] \quad (\text{by independence between } K \text{ and } M) \\ &= \frac{1}{2} \\ &= P_C(0) \\ &= P_C(1) \quad (\text{because } P_C(1) + \underbrace{P_C(0)}_{=\frac{1}{2}} = 1) \end{aligned}$$